CBSE 2027 Chemistry — Predicted Paper (Set 5 of 10)

Based on the official CBSE curriculum, recent sample paper designs, and the frequency of concepts tested in past board exams (PYQs), here is Set 5 of the predicted board paper. All questions are completely unique from previous sets and target critical, high-probability topics.

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. According to Raoult's law, the relative lowering of vapour pressure for a solution containing a non-volatile solute is equal to the: (a) mole fraction of the solvent. (b) mole fraction of the solute. (c) mass percentage of the solute. (d) molality of the solution. * **Chapter:** Solutions | **Confidence:** Very High

Q2. Given the standard reduction potentials: $E^{\circ}_{K^+/K} = -2.93\,V$, $E^{\circ}_{Ag^+/Ag} = 0.80\,V$, $E^{\circ}_{Hg^{2+}/Hg} = 0.79\,V$, and $E^{\circ}_{Mg^{2+}/Mg} = -2.37\,V$. Which of the following is the strongest reducing agent? (a) Ag (b) Mg (c) K (d) Hg * **Chapter:** Electrochemistry | **Confidence:** Very High

Q3. The half-life period ($t_{1/2}$) of a zero-order reaction is: (a) independent of the initial concentration of the reactant. (b) directly proportional to the initial concentration of the reactant. (c) inversely proportional to the initial concentration of the reactant. (d) directly proportional to the square of the initial concentration. * **Chapter:** Chemical Kinetics | **Confidence:** High

Q4. Which of the following pairs of ions possesses the exact same $3d^5$ electronic configuration? (a) Fe^{3+} and Co^{3+} (b) Cr^{3+} and Mn^{2+} (c) Fe^{3+} and Mn^{2+} (d) Mn^{3+} and Fe^{2+} * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q5. Which type of isomerism is exhibited by the coordination complex $[Co(NH_3)_5(SCN)]^{2+}$ and $[Co(NH_3)_5(NCS)]^{2+}$? (a) Ionization isomerism (b) Linkage isomerism (c) Coordination isomerism (d) Solvate isomerism * **Chapter:** Coordination Compounds | **Confidence:** Very High

Q6. The most preferred reagent for converting ethanol to chloroethane is: (a) PCl_3 (b) PCl_5 (c) $SOCl_2$ in the presence of pyridine (d) HCl and anhydrous $ZnCl_2$ * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q7. The acid-catalysed hydration of propene yields which of the following as the major product? (a) Propan-1-ol (b) Propan-2-ol (c) Propane (d) Propanal * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** High

Q8. Arrange the following in the increasing order of their acidic strength: Acetic acid, Chloroacetic acid, Fluoroacetic acid, Trichloroacetic acid. (a) Acetic acid < Chloroacetic acid < Fluoroacetic acid < Trichloroacetic acid (b) Acetic acid < Fluoroacetic acid < Chloroacetic acid <

Trichloroacetic acid (c) Trichloroacetic acid < Fluoroacetic acid < Chloroacetic acid < Acetic acid (d) Chloroacetic acid < Fluoroacetic acid < Acetic acid < Trichloroacetic acid * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q9. The correct increasing order of basic strength for the following amines in an **aqueous solution** is: (a) $NH_3 < CH_3NH_2 < (CH_3)_2NH < (CH_3)_3N$ (b) $NH_3 < (CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$ (c) $(CH_3)_3N < (CH_3)_2NH < CH_3NH_2 < NH_3$ (d) $NH_3 < (CH_3)_2NH < CH_3NH_2 < (CH_3)_3N$ * **Chapter:** Amines | **Confidence:** High

Q10. The hydrolysis of sucrose (cane sugar) yields: (a) Two molecules of α-D-glucose (b) One molecule of α-D-glucose and one molecule of β-D-fructose (c) Two molecules of β-D-fructose (d) One molecule of β-D-glucose and one molecule of α-D-fructose * **Chapter:** Biomolecules | **Confidence:** Very High

Q11. Grignard reagents ($R - Mg - X$) must be prepared under strictly anhydrous conditions because: (a) they are highly volatile. (b) they decompose into alkenes in the presence of moisture. (c) they react with water to form alkanes. (d) they react with water to form alcohols. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q12. What is the oxidation state of the central metal atom in the complex $K_3[Fe(CN)_6]$? (a) +2 (b) +3 (c) +4 (d) +6 * **Chapter:** Coordination Compounds | **Confidence:** High

Directions for Q13 to Q16: Two statements are given—one labeled Assertion (A) and the other Reason (R). Select the correct answer from the codes (a), (b), (c), and (d). *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q13. Assertion (A): Two isotonic solutions separated by a semi-permeable membrane do not show the phenomenon of osmosis. **Reason (R):** Isotonic solutions have the exact same molar concentration and osmotic pressure at a given temperature. * **Chapter:** Solutions | **Confidence:** Very High

Q14. Assertion (A): At equilibrium, the standard cell potential (E°_{cell}) of a galvanic cell becomes zero. **Reason (R):** At equilibrium, the reaction quotient (Q_c) becomes equal to the equilibrium constant (K_c). * **Chapter:** Electrochemistry | **Confidence:** Medium

Q15. Assertion (A): The rate of a chemical reaction generally doubles for every 10°C rise in temperature. **Reason (R):** The number of collisions between reactant molecules directly doubles for every 10°C rise in temperature. * **Chapter:** Chemical Kinetics | **Confidence:** High

Q16. Assertion (A): Amino acids behave like salts rather than simple amines or carboxylic acids. **Reason (R):** In aqueous solution, the carboxyl group loses a proton and the amino group accepts a proton, forming a dipolar Zwitterion. * **Chapter:** Biomolecules | **Confidence:** Very High

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SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. A solution contains 5.85 g of $NaCl$ (Molar mass = $58.5\, g\, mol^{-1}$) dissolved in 90 g of water (Molar mass = $18\, g\, mol^{-1}$). Calculate the mole fraction of $NaCl$ in the solution. * **Chapter:** Solutions | **Confidence:** High

Q18. Differentiate between Homoleptic and Heteroleptic coordination complexes with the help of one specific example for each. * **Chapter:** Coordination Compounds | **Confidence:** High

Q19. Write the chemical equation for the synthesis of primary amines via the Hoffmann Bromamide Degradation reaction. Why does this reaction yield an amine with one carbon atom less than the parent amide? * **Chapter:** Amines | **Confidence:** Very High

Q20. Briefly explain the Dow's Process for the industrial preparation of phenol from chlorobenzene, specifying the exact temperature and pressure conditions required. * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q21. Define the term "Lanthanoid Contraction". State one of its major consequences on the properties of the post-lanthanoid transition elements (e.g., Zr and Hf). * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

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SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. A solution containing 18 g of an unknown non-volatile solute dissolved in 200 g of water boils at 100.20 °C. Calculate the molar mass of the unknown solute. (Given: Boiling point of pure water = 100 °C, K_b for water = $0.52\, K\, kg\, mol^{-1}$). * **Chapter:** Solutions | **Confidence:** Very High

Q23. Using the initial rate method, determine the order of the reaction with respect to A and B, and find the overall order of the reaction from the following data for the reaction $A + B \rightarrow Products$: * Experiment 1: [A] = 0.1 M, [B] = 0.1 M, Initial Rate = $2.0 \times 10^{-3}\, M\, s^{-1}$ * Experiment 2: [A] = 0.2 M, [B] = 0.1 M, Initial Rate = $4.0 \times 10^{-3}\, M\, s^{-1}$ * Experiment 3: [A] = 0.2 M, [B] = 0.2 M, Initial Rate = $1.6 \times 10^{-2}\, M\, s^{-1}$ * **Chapter:** Chemical Kinetics | **Confidence:** High

Q24. Give distinct chemical tests (along with their observations) to distinguish between the following pairs of organic compounds: (i) Formic acid and Acetic acid (ii) Methanol and Ethanol (iii) Acetophenone and Benzophenone * **Chapter:** Organic Chemistry (Mixed) | **Confidence:** Very High

Q25. Answer the following questions regarding haloarenes: (i) Why do haloarenes undergo electrophilic substitution reactions more slowly compared to plain benzene? (ii) Why does the presence of an electron-withdrawing group (like $-NO_2$) at the ortho and para positions heavily increase the reactivity of haloarenes towards nucleophilic substitution? * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

Q26. (i) Draw the pyranose ring structure of α-D-glucopyranose. (1 Mark) (ii) What is a glycosidic linkage? (1 Mark) (iii) Write one primary difference between a nucleoside and a nucleotide. (1 Mark) * **Chapter:** Biomolecules | **Confidence:** High

Q27. Complete the following reactions involving d- and f-block elements: (i) $2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta}$? (ii) $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow$? (iii) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow$? * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

Q28. How will you bring about the following conversions in not more than two steps? (i) Aniline to Iodobenzene (ii) Toluene to Benzoic acid (iii) Ethanenitrile to Ethanamine * **Chapter:** Organic Chemistry (Mixed) | **Confidence:** Very High

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SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Basicity of Amines and Solvation Effects Amines act as Lewis bases due to the presence of an unshared pair of electrons on the nitrogen atom. In the gaseous phase, the basicity of aliphatic amines strictly follows the inductive effect (+I effect) of the alkyl groups, making tertiary amines the most basic. However, in an aqueous solution, the basicity order is altered because it depends not only on the +I effect but also on the solvation of the substituted ammonium cation by water molecules and steric hindrance. For methylamines in aqueous solution, the order is secondary > primary > tertiary > ammonia. **(i)** Identify the most basic amine in the gaseous phase among CH_3NH_2, $(CH_3)_2NH$, and $(CH_3)_3N$. (1 Mark) **(ii)** Explain why tertiary methylamine is less basic than primary methylamine in an aqueous solution despite having three electron-donating methyl groups. (1 Mark) **(iii)** Why is aniline significantly less basic than cyclohexylamine? Draw resonance structures to support your answer. (2 Marks) * **Chapter:** Amines | **Confidence:** Very High

Q30. Case Study 2: Valence Bond Theory and Magnetic Properties Valence Bond Theory (VBT) is used to describe the bonding, hybridization, and magnetic properties of coordination complexes. The ligands donate electron pairs into the empty hybridized orbitals of the central metal ion. Depending on the strength of the ligand, the $(n-1)d$ inner orbitals or the nd outer orbitals are utilized. Strong field ligands cause the pairing of unpaired electrons in the metal's d-orbitals, forming low-spin (inner orbital) complexes, whereas weak field ligands do not cause pairing, forming high-spin (outer orbital) complexes. **(i)** What is the shape of a complex if its central metal ion exhibits dsp^2 hybridization? (1 Mark) **(ii)** Out of NH_3 and F^-, which acts as a weak field ligand causing no electron pairing? (1 Mark) **(iii)** Using VBT, predict the hybridization and the magnetic behaviour (paramagnetic or diamagnetic) of the complex $[Fe(CN)_6]^{4-}$. (Given atomic number of Fe = 26). (2 Marks) * **Chapter:** Coordination Compounds | **Confidence:** High

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SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Organic Compounds: Identification & Structure An organic compound **(A)** with the molecular formula $C_5H_{10}O_2$ is subjected to acidic hydrolysis, producing a carboxylic acid

(B) and an alcohol **(C)**. When alcohol **(C)** is subjected to oxidation using acidified $K_2Cr_2O_7$, it produces the exact same carboxylic acid **(B)**. (i) Identify compounds **(A)**, **(B)**, and **(C)**. Provide their IUPAC names. (3 Marks) (ii) Write the balanced chemical equation for the acidic hydrolysis of compound **(A)**. (1 Mark) (iii) Alcohol **(C)** reacts with iodine and sodium hydroxide. Will it yield a yellow precipitate? Justify. (1 Mark) * **OR (B)** Account for the following observations with chemical reasoning: (i) Carboxylic acids do not give the characteristic nucleophilic addition reactions of the carbonyl group (e.g., they do not form 2,4-DNP derivatives). (2 Marks) (ii) The pK_a value of chloroacetic acid is lower than that of acetic acid. (1 Mark) (iii) Aldehydes are generally more reactive than ketones towards nucleophilic addition reactions. (Provide two distinct reasons). (2 Marks) * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** Very High

Q32. (A) Electrochemistry: Kohlrausch's Law and Conductivity (i) State Kohlrausch's Law of independent migration of ions. (1 Mark) (ii) The conductivity (κ) of a $0.001028\,M$ solution of acetic acid is $4.95 \times 10^{-5}\,S\,cm^{-1}$. Calculate its molar conductivity (Λ_m). (2 Marks) (iii) If the limiting molar conductivities (Λ_m°) for H^+ and CH_3COO^- are $349.6\,S\,cm^2\,mol^{-1}$ and $40.9\,S\,cm^2\,mol^{-1}$ respectively, calculate the degree of dissociation (α) of the acetic acid solution. (2 Marks) * **OR (B)** (i) Differentiate between primary and secondary batteries. Give one standard example of each. (2 Marks) (ii) Write the Nernst equation for a generic reaction $aA + bB \rightarrow cC + dD$. (1 Mark) (iii) The standard electrode potentials are $E^\circ_{Cu^{2+}/Cu} = 0.34\,V$ and $E^\circ_{Zn^{2+}/Zn} = -0.76\,V$. Calculate the standard Gibbs free energy change (ΔG°) for the Daniell cell operating at standard conditions. (Given: $1F = 96500\,C\,mol^{-1}$). (2 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High

Q33. (A) Chemical Kinetics: Collision Theory and Arrhenius Equation (i) What are the two main requirements (criteria) according to Collision Theory that must be satisfied for a collision between reactant molecules to yield products? (2 Marks) (ii) The specific rate constant for the decomposition of a substance is $1.5 \times 10^{-4}\,s^{-1}$ at 300 K. If the activation energy (E_a) for the reaction is $50\,kJ\,mol^{-1}$, calculate the rate constant at 320 K. (Given: $R = 8.314\,J\,K^{-1}\,mol^{-1}$, antilog values can be kept in logarithmic form if exact calculation is tedious, but show the full Arrhenius equation setup). (3 Marks) * **OR (B)** (i) Show that for a first-order reaction, the time required for 99.9% completion is roughly 10 times its half-life ($t_{1/2}$). (3 Marks) (ii) Define the term "temperature coefficient" of a chemical reaction. (1 Mark) (iii) How does the presence of a positive catalyst affect the activation energy and the enthalpy (ΔH) of the reaction? (1 Mark) * **Chapter:** Chemical Kinetics | **Confidence:** Very High

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CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.